Imaginary Vector Algebra:
A Rigorous Framework for Unifying State and Phase

Pearl Bipin Pulickal
Department of Electronics and Communication Engineering
National Institute of Technology, Goa

August 13, 2025

Abstract

This paper develops Imaginary Vector Algebra (IVA), a rigorous mathematical framework for unifying the description of static vector quantities and dynamic oscillatory phenomena. While established formalisms such as Complex Vector Spaces ($\mathbb{C}^n$) and Geometric Algebra provide comprehensive tools, IVA offers a pedagogically distinct approach by explicitly separating a vector entity into a real 'state' component and an imaginary 'phase' component. We define an Imaginary Vector as $\mathbf{Z} = \vec{a} + i\vec{b}$, where $\vec{a}, \vec{b} \in \mathbb{R}^n$. We establish its properties, including a conjugate-symmetric inner product and a formalism for linear operators. The framework's utility is demonstrated through a detailed numerical analysis of an RLC circuit and the visualization of a quantum bit (qubit). By preserving directional vector information while encoding phase, IVA presents an intuitive and powerful tool for analysis and education in engineering and physics.

Contents

1 Introduction

In the physical sciences, a methodological gap often exists between the description of static and dynamic systems. Static properties like displacement or force are adeptly handled by real vectors in $\mathbb{R}^n$, such as $\vec{F}$, where the direction and magnitude are explicit. Conversely, dynamic systems involving oscillation or rotation are typically modeled using complex numbers, where a phasor $Ae^{i\omega t}$ elegantly captures amplitude and phase.

While powerful, this dual approach can obscure the underlying unity of a system. A voltage in a circuit, for example, is both a potential difference with a direction and an oscillating quantity with a phase. Formalisms like Complex Vector Spaces ($\mathbb{C}^n$) and the more encompassing Geometric Algebra [1] provide mathematically complete descriptions. However, their abstraction can present a steep learning curve.

This paper develops **Imaginary Vector Algebra (IVA)** as a specialized framework designed for pedagogical clarity and intuitive physical insight. It formalizes an object—the Imaginary Vector—that explicitly contains both a real **state vector** and an orthogonal imaginary **phase vector**. We aim to:

1. Establish a mathematically rigorous algebra for IVA, including proofs of its inner product properties and a definition for linear operators.
2. Demonstrate its practical utility by solving a numerical RLC circuit problem.
3. Provide clear visualizations to make the concepts intuitive.
4. Situate IVA within the context of existing theories, highlighting its unique advantages and limitations.

Throughout this paper, unit vectors in $\mathbb{R}^3$ are denoted by a hat, e.g., $\hat{x}, \hat{y}, \hat{z}$.

2 The Rigorous Formalism of Imaginary Vector Algebra

2.1 Definitions and Basic Operations

Definition 2.1 (Imaginary Vector Space)**.** *The **Imaginary Vector Space**, denoted $\mathbb{V}_I^n$, is the set of all objects $\mathbf{Z}$ of the form $\mathbf{Z} = \vec{a} + i\vec{b}$, where $\vec{a}, \vec{b} \in \mathbb{R}^n$ and $i^2 = -1$. We call $\vec{a} = \mathrm{Re}(\mathbf{Z})$ the **state vector** and $\vec{b} = \mathrm{Im}(\mathbf{Z})$ the **phase vector**.*

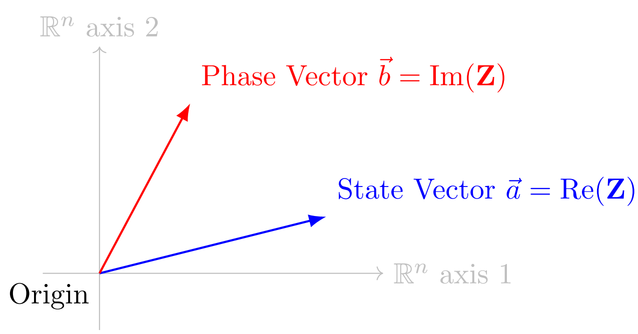

Figure 1: Conceptual visualization of an Imaginary Vector $\mathbf{Z}$, composed of a real state vector $\vec{a}$ and an orthogonal imaginary phase vector $\vec{b}$.

Addition and scalar multiplication follow naturally, forming a vector space over $\mathbb{C}$.

2.2 The IVA Inner Product

The inner product is defined to capture physical interactions, such as work or power.

Definition 2.2 (IVA Inner Product). *The inner product of* $\mathbf{Z}_1 = \vec{a}_1 + i\vec{b}_1$ *and* $\mathbf{Z}_2 = \vec{a}_2 + i\vec{b}_2$ *is a complex scalar given by:*

$$\langle \mathbf{Z}_1, \mathbf{Z}_2 \rangle = (\vec{a}_1 - i\vec{b}_1) \cdot (\vec{a}_2 + i\vec{b}_2) = (\vec{a}_1 \cdot \vec{a}_2 + \vec{b}_1 \cdot \vec{b}_2) + i(\vec{a}_1 \cdot \vec{b}_2 - \vec{b}_1 \cdot \vec{a}_2) \tag{1}$$

where '$\cdot$' is the standard Euclidean dot product. This definition is a natural extension of the Hermitian inner product on $\mathbb{C}^n$.

Proposition 2.1 (Properties of the Inner Product). *The IVA inner product satisfies conjugate symmetry and linearity in its second argument.*

Proof. **1. Conjugate Symmetry:** $\langle \mathbf{Z}_2, \mathbf{Z}_1 \rangle = \overline{\langle \mathbf{Z}_1, \mathbf{Z}_2 \rangle}$.

$$\begin{aligned}
\langle \mathbf{Z}_2, \mathbf{Z}_1 \rangle &= (\vec{a}_2 \cdot \vec{a}_1 + \vec{b}_2 \cdot \vec{b}_1) + i(\vec{a}_2 \cdot \vec{b}_1 - \vec{b}_2 \cdot \vec{a}_1) \\
&= (\vec{a}_1 \cdot \vec{a}_2 + \vec{b}_1 \cdot \vec{b}_2) - i(\vec{b}_1 \cdot \vec{a}_2 - \vec{a}_1 \cdot \vec{b}_2) \\
&= \overline{(\vec{a}_1 \cdot \vec{a}_2 + \vec{b}_1 \cdot \vec{b}_2) + i(\vec{a}_1 \cdot \vec{b}_2 - \vec{b}_1 \cdot \vec{a}_2)} = \overline{\langle \mathbf{Z}_1, \mathbf{Z}_2 \rangle}
\end{aligned}$$

2. Linearity in the second argument is inherited directly from the linearity of the dot product. □

Corollary 2.1 (The IVA Norm). *The norm-squared of a vector* $\mathbf{Z} = \vec{a} + i\vec{b}$ *is a real, positive semi-definite scalar.*

$$\|\mathbf{Z}\|^2 = \langle \mathbf{Z}, \mathbf{Z} \rangle = |\vec{a}|^2 + |\vec{b}|^2 \tag{2}$$

Proof. Using Eq. 1, let $\mathbf{Z}_1 = \mathbf{Z}_2 = \mathbf{Z}$. $\|\mathbf{Z}\|^2 = (\vec{a} \cdot \vec{a} + \vec{b} \cdot \vec{b}) + i(\vec{a} \cdot \vec{b} - \vec{b} \cdot \vec{a})$. Since the dot product is commutative ($\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$), the imaginary part vanishes, leaving $\|\mathbf{Z}\|^2 = |\vec{a}|^2 + |\vec{b}|^2$. □

2.3 Linear Operators in IVA

An IVA linear operator $\hat{O}$ is a map $\hat{O} : \mathbb{V}_I^n \to \mathbb{V}_I^n$. It can be represented by a 2×2 block matrix of real-valued linear operators.

$$\hat{O} = \begin{pmatrix} O_{aa} & O_{ab} \\ O_{ba} & O_{bb} \end{pmatrix} \tag{3}$$

Its action on $\mathbf{Z} = \vec{a} + i\vec{b}$ is defined as:

$$\hat{O}\mathbf{Z} = (O_{aa}\vec{a} + O_{ab}\vec{b}) + i(O_{ba}\vec{a} + O_{bb}\vec{b}) \tag{4}$$

This allows us to model complex interactions, such as impedance, where the state and phase components influence each other.

3 Numerical Example: RLC Circuit Analysis with IVA

Let's analyze a series RLC circuit with the following parameters:

- Resistor: $R = 10\,\Omega$
- Inductor: $L = 20\,\text{mH}$
- Capacitor: $C = 500\,\mu\text{F}$
- Voltage Source: $V(t) = 10\cos(100t)\,\text{V}$, with angular frequency $\omega = 100\,\text{rad/s}$.

Let the current flow in the $\hat{k}$ direction. We want to find the steady-state current $I(t)$.

1. IVA Representation. We represent the voltage source as an IVA vector where the state part is the instantaneous voltage and the phase part is its time-integral (related to flux linkage).

$$\mathbf{V}(t) = 10\cos(100t)\hat{k} + i\int(10\cos(100t)\hat{k})dt = 10\cos(100t)\hat{k} + i\frac{10}{100}\sin(100t)\hat{k} \tag{5}$$

This is effectively the real and imaginary parts of the phasor $10e^{i100t}$ mapped to IVA state/phase vectors. Let the unknown current be $\mathbf{I} = I_{state} + iI_{phase}$.

2. IVA Impedance Operators. The impedance of each component is an IVA operator.

- Resistor: $V_R = RI \implies \hat{Z}_R = \begin{pmatrix} R & 0 \\ 0 & R \end{pmatrix}$
- Inductor: $V_L = L\frac{dI}{dt} \implies \hat{Z}_L = \begin{pmatrix} 0 & -\omega L \\ \omega L & 0 \end{pmatrix}$
- Capacitor: $V_C = \frac{1}{C}\int Idt \implies \hat{Z}_C = \begin{pmatrix} 0 & 1/(\omega C) \\ -1/(\omega C) & 0 \end{pmatrix}$

The total impedance operator is $\hat{Z}_{total} = \hat{Z}_R + \hat{Z}_L + \hat{Z}_C$. Plugging in values: $\omega L = 100 \cdot 0.02 = 2\,\Omega$ and $1/(\omega C) = 1/(100 \cdot 500 \cdot 10^{-6}) = 20\,\Omega$.

$$\hat{Z}_{total} = \begin{pmatrix} 10 & 0 \\ 0 & 10 \end{pmatrix} + \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 20 \\ -20 & 0 \end{pmatrix} = \begin{pmatrix} 10 & 18 \\ -18 & 10 \end{pmatrix} \tag{6}$$

3. Solving for Current. We solve the IVA Ohm's Law $\mathbf{V} = \hat{Z}_{total}\mathbf{I}$, which means $\mathbf{I} = \hat{Z}_{total}^{-1}\mathbf{V}$. The inverse operator is $\hat{Z}_{total}^{-1} = \frac{1}{10^2+18^2}\begin{pmatrix} 10 & -18 \\ 18 & 10 \end{pmatrix} = \frac{1}{424}\begin{pmatrix} 10 & -18 \\ 18 & 10 \end{pmatrix}$.

Applying this to $\mathbf{V}(t) = (10\cos\omega t)\hat{k} + i(0.1\sin\omega t)\hat{k}$:

$$\begin{aligned} \mathbf{I}(t) &= \frac{1}{424}\begin{pmatrix} 10 & -18 \\ 18 & 10 \end{pmatrix}\begin{pmatrix} 10\cos\omega t \\ 0.1\sin\omega t \end{pmatrix}\hat{k} \\ &= \frac{1}{424}\left[(100\cos\omega t - 1.8\sin\omega t) + i(180\cos\omega t + \sin\omega t)\right]\hat{k} \end{aligned}$$

The actual instantaneous current is the state vector $\mathrm{Re}(\mathbf{I}(t))$. This result is identical to traditional phasor analysis but provides a full state/phase vector at every time step.

4 Applications and Visualizations

4.1 Quantum Mechanics: Visualizing a Qubit

A qubit state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ can be mapped to $\mathbf{Z}_\psi = (\mathrm{Re}(\alpha)\hat{x} + \mathrm{Re}(\beta)\hat{y}) + i(\mathrm{Im}(\alpha)\hat{x} + \mathrm{Im}(\beta)\hat{y})$. This allows us to visualize the state in a "State-Phase Plane."

5 Discussion and Comparative Analysis

5.1 Advantages of IVA

The primary advantage of IVA is its **intuitive, geometric representation** that preserves vectorial information often lost in scalar complex analysis.

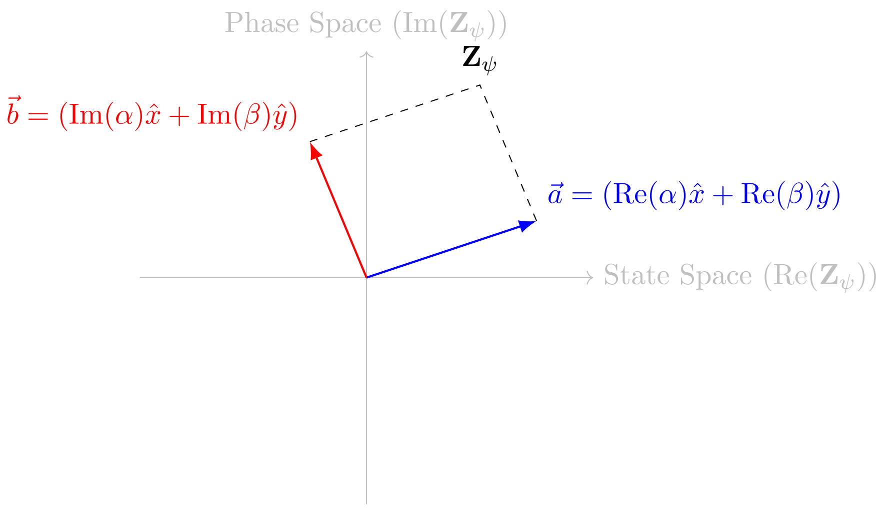

Figure 2: The State-Phase Plane for a qubit represented as an IVA vector. The total state is a combination of the real state vector and the imaginary phase vector.

Table 1: Comparison of Traditional Phasor Analysis and IVA for an AC Circuit.

Aspect	**Traditional Phasor Analysis**	**Imaginary Vector Algebra (IVA)**
Entity	Complex Scalar (V, I)	Imaginary Vector ($\mathbf{V}, \mathbf{I}$)
Direction	Implicit or lost	Explicit (e.g., in $\hat{k}$)
Impedance	Complex Scalar (Z)	Linear Operator ($\hat{Z}$)
Interpretation	Algebraic/Geometric in 2D plane	Geometric in State+Phase Space
Benefit	Computationally swift	Physically intuitive, preserves geometry

5.2 Relation to Existing Formalisms

- **Complex Vector Spaces ($\mathbb{C}^n$):** IVA is structurally isomorphic to $\mathbb{C}^n$. The key difference is interpretive: IVA enforces a conceptual split between the real and imaginary vector components, assigning them distinct physical roles (state vs. phase), which is not a required feature of $\mathbb{C}^n$.

- **Geometric Algebra (GA):** GA is a far more general and powerful framework. The rotor element $e^{i\theta}$ in GA performs rotations on vectors in a plane. IVA can be seen as a simplified subset of GA's ideas, tailored for ease of use in systems where a single phase-plane rotation is dominant.

5.3 Limitations

IVA is not a universal tool. For problems involving arbitrary 3D spatial rotations (e.g., rigid body dynamics), quaternion algebra is more direct. For high-level theoretical physics, the generality of Geometric Algebra or tensor calculus is superior. IVA's utility is centered on systems with coupled state and phase dynamics.

6 Future Work and Outlook

Calculus of IVA. A natural next step is to formalize IVA calculus. The derivative of an IVA vector $\mathbf{Z}(t) = \vec{a}(t) + i\vec{b}(t)$ is naturally defined as:

$$\frac{d\mathbf{Z}}{dt} = \frac{d\vec{a}}{dt} + i\frac{d\vec{b}}{dt} \tag{7}$$

This would allow for the formulation of differential equations entirely within IVA, describing the co-evolution of state and phase.

Concrete Research Directions.

1. **Advanced Quantum Systems:** Extend the operator formalism to represent quantum gates for multi-qubit systems and visualize entanglement within the IVA state-phase space.

2. **Computational Implementation:** Develop a Python library (using NumPy) to implement IVA objects and operators. This would create a powerful simulation tool for analyzing complex AC circuits, coupled oscillators, or introductory quantum algorithms.

3. **Electromagnetism:** Reformulate Maxwell's equations using IVA, where the electric and magnetic fields could be combined into a single Imaginary Field Vector, offering new insights into electromagnetic wave propagation.

References

[1] D. Hestenes, *Clifford Algebra to Geometric Calculus: A Unified Language for Mathematics and Physics*. Springer Science & Business Media, 2012.

[2] M. N. O. Sadiku and S. V. Kulkarni, *Elements of Electromagnetics*. Oxford University Press, 2017.

[3] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*. Cambridge University Press, 2010.

[4] L. A. Pipe, *Matrix Methods for Engineering*. Prentice-Hall, 1966.